

Mahika Gunjkar

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA

Dec 2026

Master of Information Systems Management – Business Intelligence & Data Analytics

Coursework: Intro to Deep Learning, Unstructured Data Analytics, Machine Learning for Problem Solving, Advanced Business Analytics, Statistics

University of Cincinnati, Cincinnati, OH

May 2025

Bachelor of Business Administration – Information Systems | Certificate: Business Analytics

Coursework: Data Mining, Business Intelligence, IT Architecture & Networks, Advanced App Development, Forecast Risk & Analysis

TECHNICAL SKILLS

- **Deep Learning:** PyTorch, CNNs, RNNs/LSTMs, Transformers, Diffusion Models, VAEs, OT-CFM, Classifier-Free Guidance, Metric Learning
- **Generative AI & LLMs:** Latent Diffusion, Masked DiT, RAG Architectures, Prompt Engineering, Agentic Workflows, Gemini 1.5 SDK
- **Data & Cloud Engineering:** Python, SQL/PostgreSQL, AWS (EC2, G4), Docker, Git, ETL Pipelines, Flask, MongoDB, React.js, TypeScript, JavaScript, Tailwind CSS, Chart.js, Scikit-learn, NumPy, Pandas

WORK EXPERIENCE

Research Assistant – CMU PLUS Lab | Carnegie Mellon University

May 2026 – Present

- Building a React and Python web app for middle school math teachers to track student progress and set learning goals from i-Ready data, with automated weekly reporting and AWS-hosted infrastructure.
- Engineering ETL pipelines to process and integrate multi-source student performance datasets into real-time teacher-facing dashboards, reducing manual reporting overhead significantly.

Software Development Intern | Revdau Industries

May 2024 – July 2024

- Engineered regression and clustering models in Scikit-learn achieving 90% accuracy on internal KPIs through feature engineering, cross-validation, and hyperparameter optimization.
- Built automated ETL pipelines in Python reducing manual data prep time by 25%; architected AWS (EC2, G4) data storage maintaining 99% uptime across cross-functional teams.
- Conducted large-scale EDA using SQL and Pandas, surfacing patterns across complex datasets that directly influenced product roadmap and strategic decisions.

Resource Management Group Intern | Tata Consultancy Services

May 2023 – July 2023

- Allocated 100+ employees to high-priority projects by analyzing skill matrices against demand-capacity models, optimizing utilization and identifying gaps that informed hiring and training plans.
- Designed and automated daily/weekly workforce tracking dashboards, improving reporting latency and giving senior leadership real-time visibility into resourcing decisions.

PROJECT EXPERIENCE

OT-AdaMask Flow: Time-Adaptive Masked DiT with Optimal Transport

May 2026

- Built a class-conditional latent flow Transformer (DiT-XL, 721M params) combining minibatch OT-CFM noise-data pairing with DDPM/DDIM schedulers and time-adaptive cosine token masking ($\gamma_{\max}=0.75$) to reduce training-time attention FLOPs while preserving full inference quality.
- Ran a 2×2 factorial ablation (400-epoch, Heun-40, CFG-1.75) revealing a non-additive interaction: OT+Mask achieved FID5k=14.04 vs. baseline 15.21, OT-only 15.73, and Mask-only 15.43; outperforming the additive prediction by 1.91 FID points.

End-to-End Speech Recognition | RNNs & CTC Loss

Feb 2026

- Built a sequence-to-sequence model using stacked LSTMs/GRUs with CTC Loss for phoneme prediction from raw audio, eliminating frame-level annotation requirements.
- Applied Beam Search Decoding and SpecAugment augmentation, achieving measurable Levenshtein Distance reduction on the competitive class leaderboard.

Facial Recognition & Verification System | CNNs

Feb 2026

- Trained a deep CNN for closed-set classification and open-set verification across thousands of identities using Triplet Loss and Center Loss for metric learning, achieving EER below 0.08.

Modular Neural Network Engine | NumPy from-scratch

Jan 2026

- Built a full deep learning library in NumPy with forward/backward passes for Linear, ReLU, and Softmax layers, custom Adam and RMSProp optimizers, and a gradient-checking suite validated against PyTorch autograd.